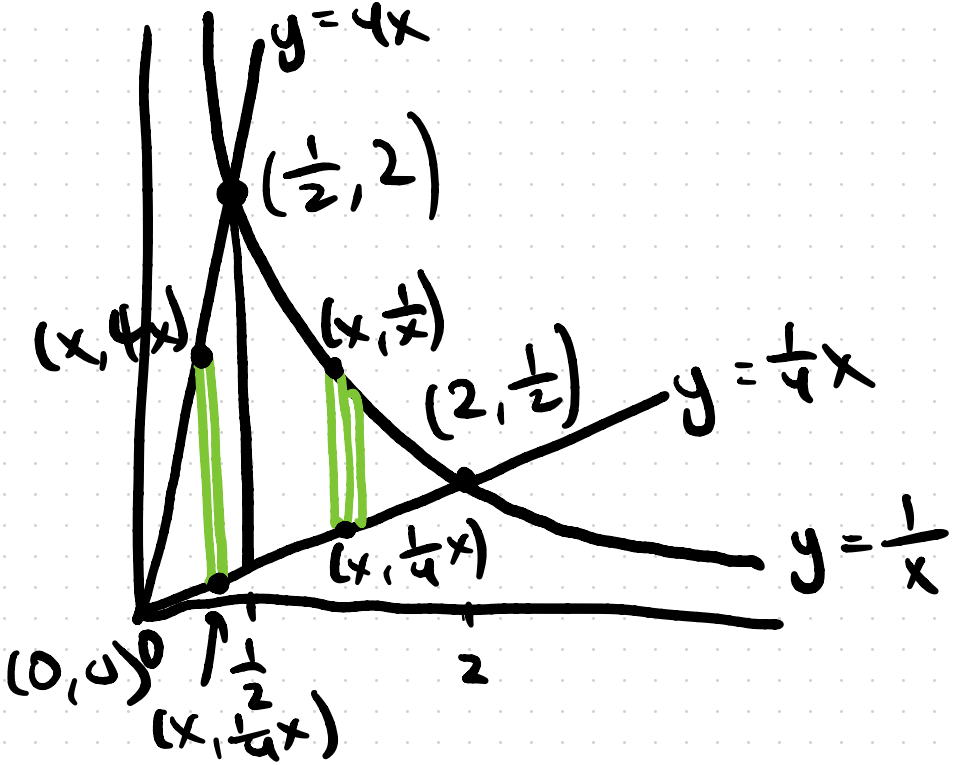
Find the area of the region bounded by $y = 4x$, $y = \frac{1}{4}x$, $y = \frac{1}{x}$.



$4x = \frac{1}{x}$

$x^2 = \frac{1}{4}$

$x = \pm\frac{1}{2}$ first quad.

$x = \frac{1}{2}$

$\frac{1}{4}x = \frac{1}{x}$

$x^2 = 4$

$x = 2$

$$\int_0^{1/2}\left(4x-\frac{1}{4}x\right)dx + \int_{1/2}^{2}\left(\frac{1}{x}-\frac{1}{4}x\right)dx$$

$$\int_0^{1/2}\frac{15}{4}x\,dx + \int_{1/2}^{2}\left(\frac{1}{x}-\frac{1}{4}x\right)dx$$

$$\frac{15}{8}x^2\Big|_0^{1/2} + \left(\ln|x| - \frac{1}{8}x^2\right)\Big|_{1/2}^{2}$$

$$\frac{15}{32} + \left(\ln 2 - \frac{1}{2}\right) - \left(\ln\frac{1}{2} - \frac{1}{32}\right)$$

$\ln 2 - \ln \frac{1}{2}$

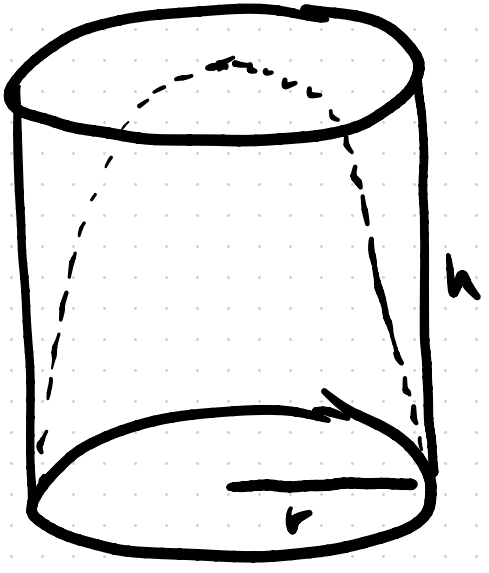
$\ln 2 - (-\ln 2)$

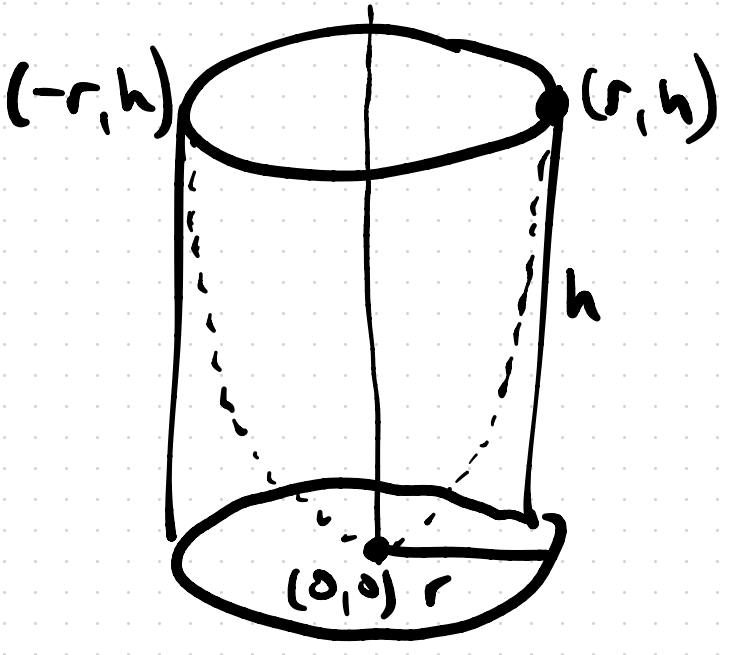
$2 \ln 2$

$\ln 4$

$\ln \frac{1}{2} = \ln 2^{-1}$

$= -\ln 2$





$y = ax^2$

Eq. of parabola is $y = ax^2 + bx + c$. Plug in the three points to form three equations, solve for a, b, c.

@ $(0,0)$ $\quad 0 = c$

@ (r,h) $\quad h = ar^2 + br$

@ $(-r,h)$ $\quad h = ar^2 - br$

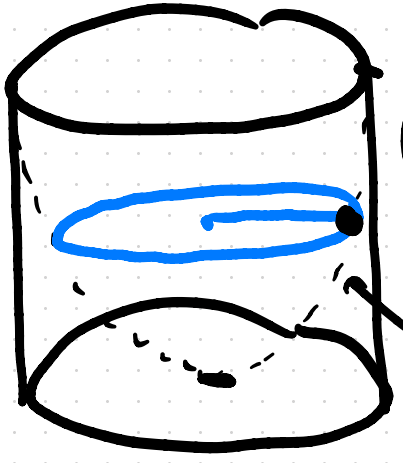
$K = K + br$

$0 = b$

$2h = 2ar^2$

$a = \frac{h}{r^2}$

$$y = \frac{h}{r^2}x^2$$



$\left(x, \frac{h}{r^2}x^2\right)$

$\left(\sqrt{\frac{yr^2}{h}}, y\right)$

$y = \frac{h}{r^2}x^2$

Integrate areas of cross sections in the y-direction.

Cross-sections are circles with radius $\sqrt{\frac{yr^2}{h}}$.

$x^2 = \frac{yr^2}{h}$

$x = \pm\sqrt{\frac{yr^2}{h}}$

Volume is

$$\int_0^h \pi\left(\sqrt{\frac{yr^2}{h}}\right)^2 dy$$

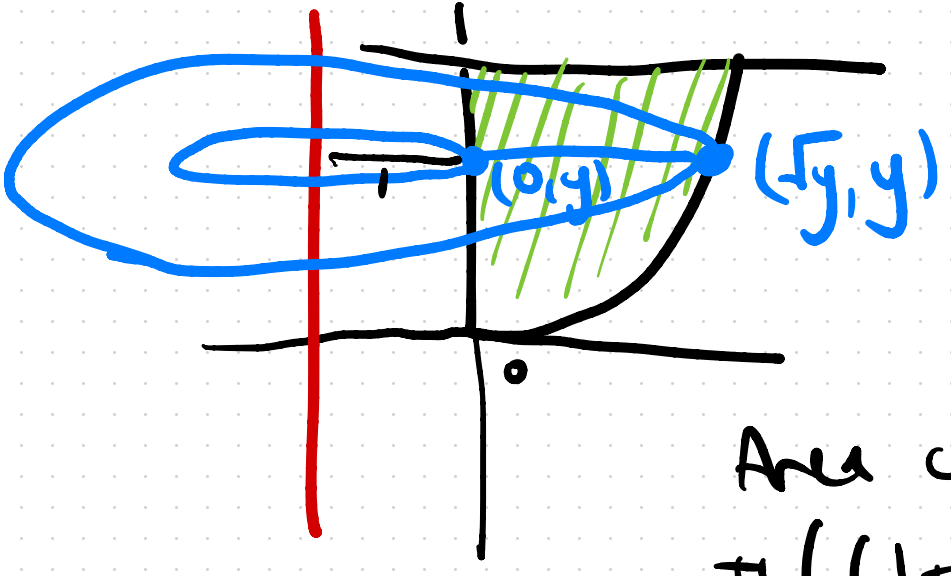
$$\int_0^h \frac{\pi r^2}{h} y \, dy$$

$$\frac{\pi r^2}{2h} y^2 \Big|_0^h = \frac{\pi r^2 h}{2}$$

Volume of cylinder: $\pi r^2 h$

Region bounded by $y = x^2$ $x = 0$ $y = 1$

about $x = -1$ $x = \pm\sqrt{y}$



Integrate in y-direction

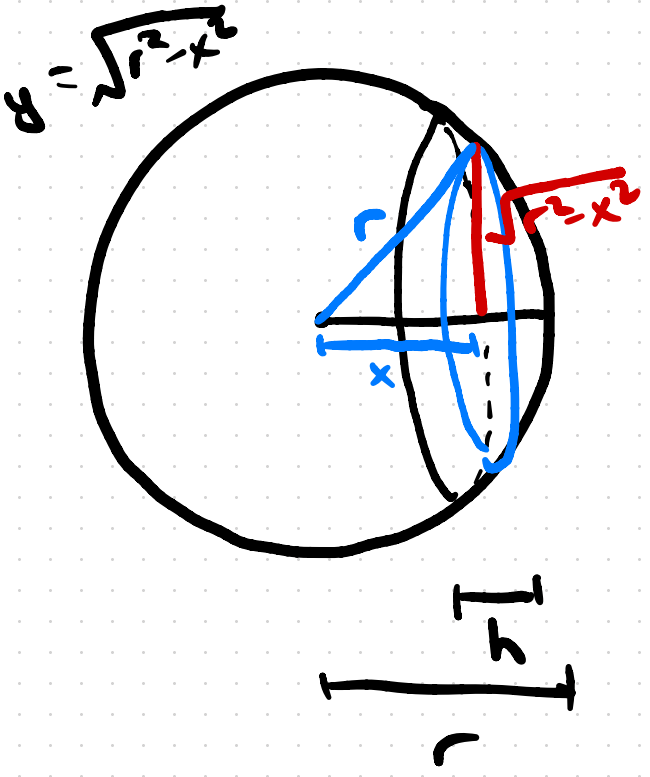
inner radius = 1

outer radius = $1+\sqrt{y}$

Area cross section

$$\pi\left((1+\sqrt{y})^2 - 1^2\right)$$

$$\int_0^1 \pi\left(\left(1+\sqrt{y}\right)^2 - 1^2\right) dy$$



Integrate in the x direction.

Cross sections are circles with radius $\sqrt{r^2 - x^2}$

Volume of cap

$$\int_{r-h}^{r} \pi \left(\sqrt{r^2 - x^2}\right)^2 dx$$

$$\pi \int_{r-h}^{r} (r^2 - x^2)\, dx$$

$$\pi\left(r^2x - \frac{1}{3}x^3\right)\Big|_{r-h}^{r}$$

$$\pi\left(r^2(r-(r-h)) - \frac{1}{3}\left(r^3 - (r-h)^3\right)\right)$$